

BUSINESS REQUIREMENTS DOCUMENT

FarmConnect

A Fair-Margin Agricultural Marketplace Platform

Pre-Harvest Auctions • B2B & Institutional Bulk Orders • Agent-Assisted Onboarding

Phase 1 Pilot Region	Tamil Nadu
Phase 1 Pilot Crop	Onion
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Executive Summary

Business Problem

A typical middleman buys onions from a farmer at roughly ₹3/kg and resells at roughly ₹30/kg — close to 90% margin, with no quality guarantee, no logistics investment, and no risk-sharing in return.

Business Opportunity

Bulk and institutional buyers — restaurants, hotels, hospitals, schools — procure on a planned schedule, days or weeks in advance. That predictability is exactly what makes a direct, quality-verified channel to farmers economically workable, without needing to solve small-basket retail delivery at all.

FarmConnect Solution

A bulk/institutional marketplace built on **pre-harvest auctions** and **B2B bulk-order matching**, supported by local agents, consolidated hub logistics, and dual-checkpoint grading.



Figure 1 — FarmConnect at a glance.

Expected Benefits

Farmers get a fair, predictable price without needing to read an app or negotiate directly. Buyers get reliable, quality-verified bulk supply at a lower cost than the traditional market, with an escrow-backed guarantee against non-delivery.

Pilot Scope

A 3-month Phase 1 pilot limited to one crop (onion) and one region (Tamil Nadu), to prove the model before wider rollout.

Snapshot

Dimension	Phase 1 Position
Pilot scope	Single crop (Onion), single region (Tamil Nadu), 3-month duration
Core transaction types	Pre-harvest auctions + B2B/institutional bulk orders only
Farmer onboarding	Agent-mediated (1 agent per 50–100 farmers) — no direct app dependency for farmers
Logistics model	Asset-light hub & spoke via partnered cold storage/warehouse operators
Revenue model	Percentage-of-transaction commission, split farmer/buyer, with a minimum floor
Target GMV (pilot)	₹10–15 Lakh across 200–300 onboarded farmers and 50–80 institutional buyers

1. Document Control

1.1 Document Purpose

This Business Requirements Document (BRD) defines the business problem, objectives, scope, stakeholders, functional and non-functional requirements, business rules, and risks associated with FarmConnect — a bulk and institutional agricultural marketplace platform. It is intended to be read by business stakeholders, product and engineering teams, and reviewers assessing the project's readiness for a Phase 1 pilot. This document describes what the business needs and why; it deliberately does not prescribe how the system will be technically built.

1.2 Version History

Version	Status	Author	Description of Change
0.1	Draft	Sahana Rajaram	Initial problem statement and concept capture

1.3 Approval Sign-off

Name	Role	Decision	Date
XXXX	Business Sponsor	Pending	
XXXX	Product Owner	Pending	
XXXX	Technical Lead	Pending	

2. Overview

2.1 Business Overview

FarmConnect is a bulk/institutional agricultural marketplace that connects farmers directly with restaurants, hotels, hospitals, schools, retailers, event caterers, and food processing companies. It operates through two transaction channels only: pre-harvest auctions and B2B/institutional bulk-order matching. It deliberately does not serve individual household consumers, as small-basket retail delivery economics are not viable at smallholder farmer scale.

2.2 Vision and Mission

Vision

Replace the exploitative middleman layer in Indian agriculture with a transparent, fair-margin platform that gives farmers a real price for their produce and gives bulk buyers reliable, quality-verified supply. FarmConnect is positioned as a value-adding intermediary, not as an elimination of intermediation altogether — the distinction lies in transparency, quality assurance, and how margin is shared.

Mission

Give every smallholder farmer — including those who cannot read, do not own a smartphone, or lack reliable network coverage — a genuine path to sell produce at a fair price, while giving bulk buyers a dependable, quality-checked supply chain at a lower cost than the traditional market.

2.3 Problem Statement (Summary)

Middlemen in the fresh produce supply chain typically buy from farmers at a fraction of resale value (for example, onions bought at approximately ₹3/kg and resold at approximately ₹30/kg), capturing roughly 90% margin with minimal corresponding service — no quality verification, no logistics investment, and no risk-

sharing. Farmers remain underpaid and cash-flow insecure, while institutional and bulk buyers pay inflated prices for inconsistent-quality produce. FarmConnect addresses this specifically for bulk and pre-planned transactions, where consolidated logistics, quality grading, and direct pricing are economically viable.

3. Scope

This section defines what is being built in Phase 1, what is explicitly excluded and why, and which design decisions are treated as locked rather than open for debate.

3.1 In Scope (Phase 1 Pilot)

- Pre-harvest auction system for farmer-listed produce (approx. 30 days before harvest)
- B2B and institutional/event bulk order matching with split fulfillment across multiple farmers
- Agent-based farmer onboarding and listing management
- Hub & spoke logistics coordination via partnered (not platform-owned) aggregation hubs
- Dual-checkpoint grading and quality verification
- Escrow-style payment and commission engine
- Guarantee fund and PMFBY crop insurance integration
- Reputation and rating system for farmers, agents, and buyers
- Multilingual (Tamil-first), voice-prompt and icon-based, agent-assisted UI
- Offline listing mode and SMS/IVR fallback channel
- Pilot scope: single crop (onion), single region (Tamil Nadu), 3-month duration

3.2 Out of Scope (with Rationale)

Excluded Item	Rationale
Retail / individual household consumer sales	Delivery cost exceeds transaction value at small-basket scale; not viable unit economics
Digital wallet	No stated problem requires a stored-value wallet; escrow-based settlement is sufficient
Subscription plans	Not aligned with the per-transaction commission model or farmer cash-flow patterns
Blockchain-based ledger	Adds complexity without solving a problem not already addressed by the escrow and audit-trail design
Generic AI recommendation engine	No demand data exists yet to support this; premature for Phase 1
IoT sensors / cold-chain hardware	Requires capital investment not justified before pilot demand is proven
Platform-owned warehouses/hubs	Deliberately partnered in Phase 1 to keep capital expenditure low until volume is proven
Separate delivery-partner role	Folded into the hub partner's logistics contract to avoid unnecessary stakeholder complexity

3.3 Locked Scope Decisions

The following decisions were deliberated and finalized during requirements gathering. They are treated as settled design constraints for this BRD, not open questions.

Area	Locked Decision
Transaction type	Pre-harvest auctions + B2B/institutional bulk orders only
Onboarding	Agent-based (1 agent per 50–100 farmers)
Connectivity	Agents work offline-first; SMS/IVR fallback for time-critical events
Logistics	Hub & spoke, asset-light (partnered hubs, not owned) in Phase 1
Revenue model	Percentage-of-transaction-value commission, split ~50/50 farmer/buyer, with a minimum floor and a reduced marginal rate for high-volume B2B
Insurance	PMFBY government scheme integration plus a platform bridge fund for payout delay
Returns/refunds	No reverse-logistics returns; graded settlement adjustment instead
Positioning	Fair-margin platform intermediary — not “middleman elimination”

4. Customer and Primary Stakeholders

This section identifies who FarmConnect serves, the core problem each stakeholder faces today, and how the platform addresses it — followed by a detailed persona and a stakeholder-management view for each group.

4.0 Stakeholders at a Glance

Stakeholder	Pain Point Today	FarmConnect Solution
Farmer	Captures a small fraction of final sale price; no price transparency	Direct auction/B2B pricing at a fair, predictable rate
Agent	No formal channel to earn from representing farmers	Commission-based role as the platform's trusted local channel
Bulk/Institutional Buyer	Inconsistent quality; no advance supply guarantee	Pre-verified, graded produce with pre-harvest commitment
Hub/Logistics Partner	Underutilized storage/consolidation capacity	Contracted volume via platform-routed consolidation
Platform Admin	No structured mechanism to arbitrate disputes at scale	Defined dispute-resolution and audit workflow

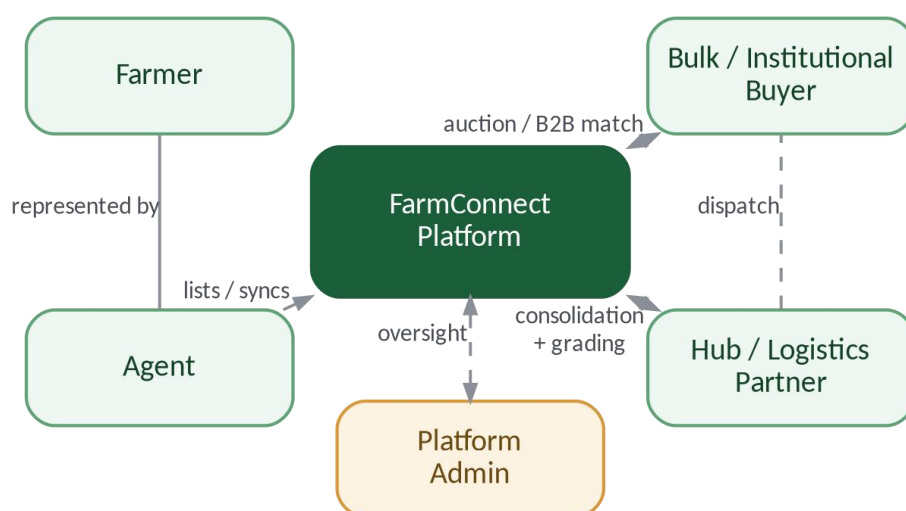


Figure 2 — FarmConnect stakeholder ecosystem: the platform sits between farmer-side and buyer-side relationships, with the Agent as the trust layer and Admin providing oversight.

4.1 Farmer (Primary Customer)

Typically a smallholder or mid-size producer growing vegetables or fruit for market sale. Often has limited formal education, may not read comfortably, and frequently lacks smartphone ownership or reliable network access. Operates on tight cash flow, where a single bad season is a serious financial threat.

Adoption depends on trust built through a local agent relationship and demonstrably better, more predictable payment than the traditional middleman route — not on the app interface itself.

4.2 Agent (Primary Customer / Channel Partner)

A literate, smartphone-owning local representative — often a cooperative staff member, agri-input dealer, or educated local resident — who manages listings on behalf of 50 to 100 farmers. The agent is the platform's actual point of digital interaction on the farmer's behalf and is the central trust relationship in the entire system.

4.3 Bulk / Institutional Buyer (Primary Customer)

Restaurants, hotels, hospitals, schools, retailers, event and catering companies, and food processing businesses that require predictable, verified volume rather than small-basket purchases. Procurement is planned days or weeks in advance, which aligns naturally with the pre-harvest auction window and B2B bulk-order matching.

4.4 Hub / Logistics Partner (Secondary Stakeholder)

An existing cold storage or warehouse operator, contracted rather than owned by FarmConnect in Phase 1. Performs consolidation of multiple farmer deliveries and an independent quality grading check before dispatch.

4.5 Platform Admin (Internal Stakeholder)

Manages dispute resolution, agent performance audits, and the guarantee fund. Acts as the arbitration point for quality mismatches and suspicious agent activity.

4.6 Stakeholder Matrix

A summary view of each stakeholder's interest in the platform's success, their influence over outcomes, and their core responsibility — a standard stakeholder-management view alongside the detailed personas above.

Stakeholder	Interest	Influence	Responsibility
Farmer	High — direct income impact	Low individually; high collectively via Agent	Deliver produce as listed and graded
Agent	High — commission-dependent	High — sole onboarding channel	Accurate listings; farmer relationship management
Bulk/Institutional Buyer	High — cost and supply reliability	Medium — demand driver	Timely payment; fair use of inspection window
Hub/Logistics Partner	Medium — contracted revenue	Medium — quality gatekeeper	Accurate grading; consolidation and dispatch
Platform Admin	High — platform integrity	High — arbitration authority	Fair dispute resolution; fraud prevention

A full RACI matrix mapping each stakeholder's involvement across key project activities is provided in **Appendix C**.

5. Goals, Objectives, and Outcome Measures

This section translates the business problem into measurable objectives and defines the specific KPIs that will determine whether the Phase 1 pilot is considered successful.

5.1 Primary Objectives

1. Increase farmer realized price per kilogram relative to the prevailing middleman rate for the pilot crop.
2. Reduce buyer-side procurement cost relative to prevailing market price for bulk/institutional purchasers.
3. Establish a trusted, agent-mediated onboarding channel that reaches farmers regardless of literacy or smartphone ownership.
4. Prove the hub & spoke logistics model keeps per-unit delivery cost economically viable at smallholder scale.
5. Demonstrate a grading and dispute-resolution mechanism sufficient to support confident sight-unseen, pre-harvest bidding.
6. Validate the two-sided commission model as sustainable platform revenue without eroding farmer or buyer trust.

5.2 Success Metrics / KPIs — Phase 1 Pilot (Onion, Tamil Nadu, 3 Months)

Metric	Target
Farmers onboarded	200–300
Active, trained agents	Determined by 1 : 50–100 farmer ratio
Institutional / B2B buyers onboarded	50–80
Gross Transaction Value (GMV) processed	₹10–15 Lakh
Auction-to-fulfillment completion rate	Target $\geq 70\%$
Grading dispute rate	Target $< 10\%$ of transactions
Farmer repeat-listing rate (2nd cycle)	Tracked as a leading trust indicator
Platform commission rate vs. prevailing middleman margin	Materially below the traditional $\sim 90\%$ margin (core proof-of-concept metric)

6. Enterprise Need / Justification

The following root-cause analysis justifies why this problem is being addressed now, and why an agent-mediated, bulk-transaction platform is the appropriate response rather than a generic e-commerce app.

6.1 Five-Why Root Cause Analysis — Farmer Side

#	Question	Answer
1	Why are farmers underpaid?	Because middlemen capture the majority of the final sale price.
2	Why can middlemen capture that much?	Because farmers have no direct market access or alternative buyer channel.
3	Why do farmers lack direct access?	Because they cannot use standard digital platforms (illiteracy, no smartphone, no network) and have no consolidated logistics option.

#	Question	Answer
4	Why has no existing solution addressed this?	Because most agri-tech platforms assume farmer self-service, and existing logistics options don't consolidate small individual harvests economically.
5	Why does this matter now?	Because a locally-trusted, agent-based onboarding model combined with consolidated hub logistics makes direct market access achievable without requiring farmers to change their digital behavior.

6.2 Five-Why Root Cause Analysis — Buyer Side

#	Question	Answer
1	Why do bulk buyers overpay for inconsistent quality?	Because every intermediary layer adds margin without adding quality assurance.
2	Why is quality inconsistent?	Because there is no standardized grading step before a transaction is finalized.
3	Why is there no grading step today?	Because traditional markets rely on in-person, at-market inspection, which doesn't scale to pre-planned bulk procurement.
4	Why can't buyers plan ahead today?	Because there is no pre-harvest commitment mechanism connecting buyer demand to farmer supply in advance.
5	Why does this matter now?	Because a pre-harvest auction with dual-checkpoint grading lets buyers commit to verified supply weeks in advance, replacing unpredictable daily spot-market purchasing.

7. Business Requirements

7.1 Business Need (Epic) and Business Requirement (User Narrative)

Each Epic below represents a core platform capability. Business requirements are expressed as user narratives in the standard “As a / I want / So that” format.

Epic 1: Pre-Harvest Auction System

- As a Farmer (via Agent), I want to list produce approximately 30 days before harvest with a minimum and maximum reserve price, so that I can secure a buyer in advance and avoid distress selling.
- As a Bulk Buyer, I want to bid on pre-harvest listings in real time, so that I can lock in verified future supply at a competitive price.

Epic 2: B2B & Institutional/Event Bulk Order Matching

- As a Bulk/Institutional Buyer, I want to post a large quantity requirement with a delivery date, so that I can source guaranteed volume for planned operations.
- As a Farmer (via Agent), I want to commit a portion of a bulk order matching my available capacity, so that I can participate in large orders I could not fulfill alone.

Epic 3: Agent Dashboard

- As an Agent, I want to manage listings for multiple farmers from a single dashboard, so that I can efficiently onboard farmers who cannot use the app directly.
- As an Agent, I want to queue listings while offline and sync when connectivity is available, so that lack of network coverage does not block my work.

Epic 4: Hub & Spoke Logistics Coordination

- As a Hub Partner, I want to view scheduled incoming deliveries from multiple farmers, so that I can plan consolidation and a single onward dispatch.
- As a Platform Admin, I want visibility into hub-level consolidation status, so that I can track fulfillment against committed orders.

Epic 5: Grading & Quality Verification

- As a Buyer, I want to see a graded photo of produce at listing time, so that I can bid with reasonable confidence despite not seeing the physical produce.
- As a Hub Partner, I want to perform an independent grading check before dispatch, so that any quality shortfall is caught and documented prior to delivery.

Epic 6: Escrow-Style Payment & Commission Engine

- As a Buyer, I want my payment held in escrow until delivery is confirmed, so that I am protected if the farmer fails to deliver as promised.
- As a Farmer, I want payment released promptly upon delivery confirmation, so that I have predictable, timely income.

Epic 7: Guarantee Fund & PMFBY Insurance Integration

- As a Farmer, I want my crop to be covered under the government PMFBY insurance scheme with a platform bridge fund for payout delay, so that a natural disaster does not leave me without any income for an extended period.

Epic 8: Reputation & Rating System

- As a Buyer, I want to view a Farmer's and Agent's rating history, so that I can make an informed bidding decision.
- As a Farmer, I want to view a Buyer's rating history, so that I can assess reliability before accepting a bid.

Epic 9: Graded Settlement Adjustment

- As a Buyer, I want a proportional payout adjustment if delivered quality does not match the promised grade, so that I am not forced to accept a full loss on a partial quality shortfall.

Epic 10: Multilingual, Agent-Assisted UI

- As a Farmer with limited literacy, I want the platform to be usable through my Agent by default, with Tamil-language, icon, and voice-prompt support where I do interact directly, so that literacy is not a barrier to participation.

Epic 11: Offline Mode & SMS/IVR Channel

- As an Agent working in low-connectivity areas, I want to collect listings offline and receive time-critical alerts via SMS/IVR, so that unreliable network access does not prevent farmer participation.

Epic 12: Security & KYC

- As a Platform Admin, I want every user type verified through KYC and every transaction encrypted, so that the platform is protected against fraud and unauthorized access.

Epic 13: Dispute Resolution Workflow

- As a Buyer, I want to flag a quality mismatch with photo/video evidence within a fixed inspection window, so that disputes are resolved fairly and quickly.
- As a Platform Admin, I want to review hub grading records against buyer-submitted evidence, so that I can make a fair settlement determination.

Future roadmap (explicitly not Phase 1 — introduced only once earned by real platform data)

- **Epic 14 — Demand Forecasting:** once FarmConnect has a real history of auction and bulk-order data, that data can advise farmers on likely next-season demand, addressing the classic overproduction/underproduction cycle. Deferred to Phase 2+ because it requires data the platform doesn't have yet at launch.
- **Epic 15 — Transaction-History-Based Credit Access:** a farmer's completed-transaction history is real, verifiable financial data that could support alternate credit scoring, in partnership with an NBFC or microfinance provider. FarmConnect would supply verified data only; the actual lending is done by the financial partner.

7.1.1 MoSCoW Prioritization

Requirement	Priority
Pre-Harvest Auction System	Must Have
B2B & Institutional Bulk Order Matching	Must Have
Agent Dashboard	Must Have
Hub & Spoke Logistics Coordination	Must Have
Grading & Quality Verification (dual checkpoint)	Must Have
Escrow-Style Payment & Commission Engine	Must Have
Dispute Resolution Workflow	Must Have
Security & KYC	Must Have
Reputation & Rating System	Should Have
Guarantee Fund + PMFBY Insurance Integration	Should Have
Offline Listing Mode + SMS/IVR Channel	Should Have
Multilingual Voice-Prompt / Icon-Based UI	Should Have
Graded Settlement Adjustment Automation	Could Have (manual admin process acceptable for pilot)
Demand Forecasting (Phase 2+)	Won't Have (this phase)
Transaction-History-Based Credit Access (Phase 3+)	Won't Have (this phase)

7.1.2 Requirements Traceability Matrix (RTM)

Req ID	Business Need	Epic	Priority	UAT Ref
BR-01	Fair price discovery for farmers	Pre-Harvest Auction System	Must	UAT-01
BR-02	Guaranteed volume for bulk buyers	B2B & Institutional Bulk Order Matching	Must	UAT-02
BR-03	Overcome illiteracy/smartphone barrier	Agent Dashboard	Must	UAT-03
BR-04	Viable delivery cost at smallholder scale	Hub & Spoke Logistics	Must	UAT-04

Req ID	Business Need	Epic	Priority	UAT Ref
BR-05	Trust in sight-unseen bidding	Grading & Quality Verification	Must	UAT-05
BR-06	Protection from buyer/farmer default	Escrow Payment & Commission Engine	Must	UAT-06
BR-07	Fair resolution of quality mismatches	Dispute Resolution Workflow	Must	UAT-07
BR-08	Fraud and unauthorized access prevention	Security & KYC	Must	UAT-08
BR-09	Trust between unfamiliar parties	Reputation & Rating System	Should	UAT-09
BR-10	Disaster income protection	Guarantee Fund + PMFBY Integration	Should	UAT-10
BR-11	Overcome connectivity barrier	Offline Mode + SMS/IVR	Should	UAT-11

7.2 User Access Levels

Role	Access Level
Farmer	No direct system access in Phase 1; represented via Agent account
Agent	Manage assigned farmer listings, view own commission, offline queue access
Bulk/Institutional Buyer	Browse auctions, place bids, post bulk orders, view own transaction history
Hub/Logistics Partner	View scheduled deliveries, submit grading records, confirm dispatch
Platform Admin	Full access: dispute resolution, agent audit, guarantee fund management, reporting

7.3 Known Interfaces and Data Sources

Interface	Purpose
Payment gateway provider	Escrow deposit and settlement processing
SMS/IVR telecom gateway	Offline / low-connectivity fallback communication
PMFBY government scheme interface	Crop insurance coverage and claims referral
KYC/identity verification service	Farmer, agent, and buyer onboarding
State agricultural regulatory data (APMC/mandi)	Compliance verification

7.4 Related Projects or Work Efforts

- **eNAM (National Agriculture Market)** — existing government e-trading platform; FarmConnect differentiates through pre-harvest auction timing, agent-based onboarding, and hub & spoke logistics.
- **State cooperative and agricultural extension programs** — potential agent recruitment channel.
- **Existing cold storage/warehouse operators** — hub partnership dependency.

7.5 Availability

The auction and bidding system must be available during active auction windows with minimal downtime, as a missed bid due to system unavailability directly affects farmer income. Target availability for the Phase 1 pilot is standard business-hours reliability, with auction closing times scheduled to avoid peak-load conflicts. 24/7 availability is not a Phase 1 requirement given limited pilot scale, but should be a target for Phase 2 scale-up.

7.6 Capacity & Performance

Parameter	Phase 1 Pilot Target
Concurrent active auctions	Up to 50
Concurrent bidders per auction	Up to 20
Registered farmers (via agents)	200–300
Registered buyers	50–80
Peak transaction volume	Aligned to onion harvest cycle in pilot region
Listing sync latency (offline agent mode)	Under 5 minutes upon connectivity restoration

7.7 Interfaces & Security

Security Requirement	Detail
Data encryption	All transactions and stored personal/financial data encrypted in transit and at rest
KYC verification	Required for all Agent, Buyer, and Hub Partner accounts prior to activation
Regulatory compliance	Compliance with India's Digital Personal Data Protection (DPDP) Act, 2023
Access control	Role-based access control aligned to the User Access Levels in Section 7.2
Audit trail	Timestamped photo records at both grading checkpoints, supporting dispute resolution and agent fraud prevention

7.8 UAT Plan

UAT Ref	Scenario	Exit Criteria
UAT-01	Agent lists produce; buyer places and wins a bid within reserve bounds	Auction closes correctly; winning bid recorded; escrow deposit triggered
UAT-02	Buyer posts bulk order; multiple farmers commit partial capacity	Order shows correct split fulfillment; total committed quantity matches requirement
UAT-03	Agent lists on behalf of an illiterate, non-smartphone-owning farmer	Listing successfully created and visible to buyers without farmer direct interaction
UAT-04	Multiple farmer deliveries consolidated at hub for single dispatch	Hub records all deliveries; single dispatch generated to buyer
UAT-05	Produce graded at listing and again at hub before dispatch	Both grading records stored and comparable for later dispute reference
UAT-06	Buyer deposit held and released only after delivery confirmation	Funds remain in escrow until confirmation step is completed
UAT-07	Buyer flags quality mismatch within inspection	Dispute created; admin able to compare hub

UAT Ref	Scenario	Exit Criteria
	window	grading vs. buyer evidence; settlement adjusted
UAT-08	Unverified user attempts to register or transact	Access denied until KYC verification is completed
UAT-09	Buyer/Farmer view counterpart's rating before transacting	Rating history correctly displayed prior to bid/commitment
UAT-10	Simulated crop-loss event triggers guarantee fund and PMFBY referral	Bridge-fund payout initiated; PMFBY claim referral generated
UAT-11	Agent queues listings offline; syncs upon reconnection	All queued listings sync correctly without data loss

8. Other Considerations

This section documents the alternatives evaluated before settling on the current design, along with the assumptions, dependencies, constraints, and risks that any implementation team should be aware of.

8.1 Alternatives Considered

Alternative	Reason Not Selected
Retail model — direct farmer-to-individual-consumer sales	Delivery cost per small basket exceeds transaction value; unfavorable unit economics at smallholder scale
Rely solely on the existing government eNAM platform rather than build a new one	eNAM operates at the mandi level and does not offer pre-harvest auction timing, agent-based onboarding, or consolidated hub & spoke logistics for smallholders
Farmer self-service app without an agent layer	Excludes a large share of farmers due to illiteracy and lack of smartphone/network access
Platform-owned warehouses from launch	Requires capital investment not justified before Phase 1 demand is proven; asset-light partnership model chosen instead
Proprietary crop insurance product	Duplicates existing government PMFBY coverage unnecessarily; integration plus a bridge fund addresses the real gap (payout delay) more efficiently

8.2 Assumptions

- Local agents can be recruited and trained within the pilot region ahead of the harvest cycle.
- At least one cold storage/warehouse operator per pilot sub-region is willing to enter a hub partnership.
- Sufficient bulk/institutional buyers (restaurants, hotels, hospitals, schools) exist within reasonable delivery range of the pilot region to generate demand.
- Farmers are willing to commit a portion of their harvest to auction 30 days in advance, retaining the remainder for traditional market fallback.
- Mobile network coverage, while inconsistent, is available intermittently enough to support an offline-sync agent model.

8.3 Dependencies

- Payment gateway provider capable of supporting escrow-style holds and delayed release.
- SMS/IVR telecom partner for low-connectivity fallback communication.

- PMFBY government scheme integration or referral pathway.
- Cold storage/warehouse operator partnership agreements executed prior to pilot launch.
- State-level regulatory clarity on direct farmer-to-buyer sale relative to APMC/mandi rules.

8.4 Constraints

- Phase 1 is limited to a single crop (onion) and single region (Tamil Nadu) to control pilot risk.
- No platform-owned logistics infrastructure in Phase 1; entirely dependent on partnered hub availability.
- Limited farmer digital literacy constrains any feature requiring direct farmer app interaction; the agent layer is a hard requirement, not an optional convenience.
- Budget and timeline for Phase 1 are scoped to a 3-month pilot window.

8.5 Business Risks and Mitigation

Risk	Mitigation
Buyer backs out after winning an auction	Escrow deposit forfeiture, partially compensating the farmer
Farmer delivers short of promised quantity/quality	Scaled payout penalty within an agreed tolerance threshold
Quality dispute at delivery	Dual grading checkpoints, fixed inspection window, and graded settlement adjustment
Agent misreports produce or colludes with a buyer	Timestamped photo audit trail at hub, periodic spot-checks, rating-threshold flagging
Natural disaster destroys a pre-sold crop	PMFBY insurance integration plus platform bridge fund for payout delay
Regulatory uncertainty around bypassing state mandi/APMC rules	Legal review required per state prior to rollout; treated as an explicit open risk
Buyer-side acquisition lags farmer-side onboarding	Prioritize securing anchor institutional buyers (hospitals, hotels with steady procurement) before scaling farmer onboarding in a given sub-region
Hub partner capacity insufficient for consolidation volume	Multiple hub partnerships per sub-region; capacity confirmed contractually before pilot launch

9. Appendices

Supporting reference material — process models, the RACI matrix, non-functional requirements, UI principles, and worked examples — that supplements the core requirements defined in Sections 1–8.

Appendix A — References

- Pradhan Mantri Fasal Bima Yojana (PMFBY) — Government of India crop insurance scheme
- Digital Personal Data Protection (DPDP) Act, 2023 — India
- eNAM (National Agriculture Market) — Government of India e-trading platform
- FAO and World Bank literature on smallholder farmer market access and intermediary margin structures
- State Agricultural Produce Market Committee (APMC) regulations — Tamil Nadu

Appendix B — Process Models (AS-IS and TO-BE)

AS-IS: Traditional Middleman-Led Supply Chain

Farmer → Local Middleman (buys at low fixed price) → Wholesale/Mandi Market → Retailer/Institutional Buyer → End Consumer. Price transparency is minimal at each step; the farmer has no visibility into final resale value and no leverage to negotiate.

TO-BE: FarmConnect Auction & B2B Flow

Farmer (via Agent) → Pre-Harvest Listing with Grading Photo → Real-Time Auction / B2B Bulk Order Match → Escrow Deposit by Buyer → Harvest → Hub Consolidation & Independent Grading → Single Dispatch to Buyer → Buyer Inspection Window → Settlement (full or grading-adjusted) → Balance Payout to Farmer. Price discovery is transparent and direct; both farmer and buyer see the same transaction terms.

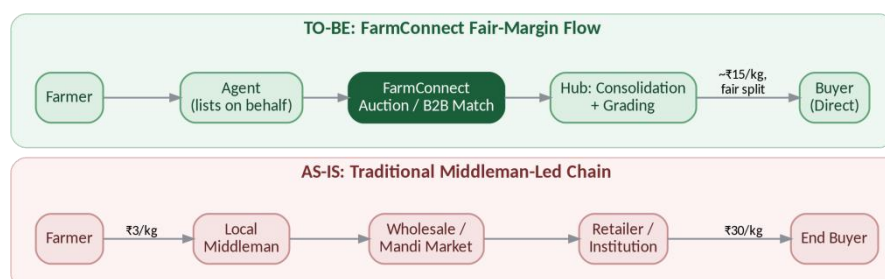


Figure 3 — AS-IS vs. TO-BE: the traditional middleman-led chain compared against the FarmConnect fair-margin flow.

End-to-End Order Lifecycle

The diagram below traces a single transaction from initial listing through to final farmer payout, spanning both the auction/B2B commitment stage and the physical fulfillment stage.

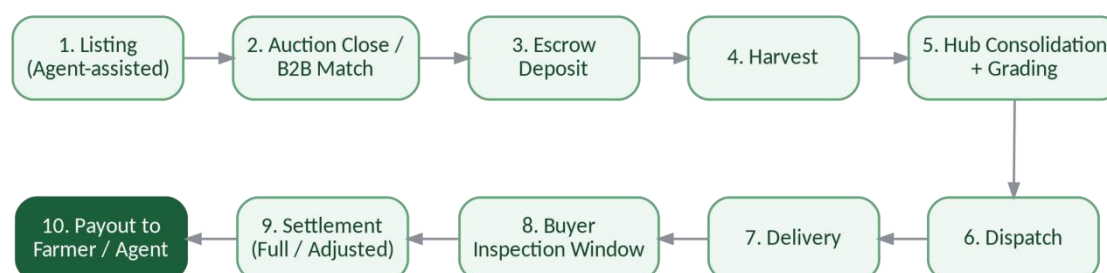


Figure 4 — FarmConnect order lifecycle, from agent-assisted listing to balance payout.

Appendix C — Stakeholders, Users, and Workgroups (RACI Matrix)

Activity	Farmer	Agent	Buyer	Hub Partner	Admin
Produce Listing	I	R/A	–	–	C
Auction Bidding	–	I	R/A	–	C
B2B Order Posting	–	I	R/A	–	C
Farmer Capacity Commitment	A	R	I	–	C
Hub Consolidation & Grading	I	I	I	R/A	C
Escrow Deposit & Release	I	I	A	C	R
Dispute Resolution	I	C	R	C	A
Agent Fraud Audit	–	R	–	C	A
Guarantee Fund / Insurance Claim	R	C	–	–	A

R = Responsible, A = Accountable, C = Consulted, I = Informed

Appendix D — Enterprise Requirements

- Integration pathway with PMFBY for crop insurance coverage and claims referral
- Compliance alignment with applicable state APMC/mandi regulations prior to regional rollout
- Alignment with India's DPDP Act, 2023 for handling of farmer, agent, and buyer personal data
- Scalability path from single-crop/single-region pilot to multi-crop, multi-region operation in Phase 2/3

Appendix E — Non-Functional Requirements

Category	Requirement
Usability	Agent-assisted-first design; Tamil-first multilingual support; icon and voice-prompt navigation for direct farmer interaction
Reliability	Offline-first agent listing mode with sync-on-reconnect; SMS/IVR fallback for time-critical events
Security	Encrypted transactions and data at rest; KYC verification for all non-farmer account types; role-based access control
Data Privacy	Full compliance with India's DPDP Act, 2023
Scalability	Architecture must support expansion from single-crop/single-region pilot to multiple crops and regions without redesign
Performance	Auction bidding must reflect near real-time updates during active bidding windows

Appendix F — User Interface / User-Centered Design Principles

- Agent-assisted flow is the primary path, not a fallback — the interface must support an agent managing many farmer accounts efficiently, not assume farmer self-service.
- Tamil-first language support for the Phase 1 pilot region, with multilingual extensibility for future regions.

- Icon-based and voice-prompt navigation to reduce reliance on text literacy wherever farmers or agents interact directly.
- Buyer-facing interface should prioritize clarity of grading information and delivery timelines over decorative browsing features, consistent with the bulk/institutional buyer's planning-oriented procurement behavior.

Appendix G — Acronyms and Abbreviations

Acronym	Meaning
BRD	Business Requirements Document
RACI	Responsible, Accountable, Consulted, Informed
MoSCoW	Must have, Should have, Could have, Won't have (prioritization framework)
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
KYC	Know Your Customer
GMV	Gross Merchandise/Transaction Value
PMFBY	Pradhan Mantri Fasal Bima Yojana (Government crop insurance scheme)
DPDP Act	Digital Personal Data Protection Act, 2023 (India)
APMC	Agricultural Produce Market Committee
eNAM	National Agriculture Market (Government of India platform)
NBFC	Non-Banking Financial Company
NFR	Non-Functional Requirement
SLA	Service Level Agreement

Appendix H — Illustrative Revenue Example & Sample Walkthroughs

Illustrative Revenue Example (not a literal universal formula)

On a fair market price of approximately ₹15/kg for onions, compared to a traditional middleman margin of approximately ₹27/kg, a total platform commission of approximately ₹10/kg (split ₹5 from the farmer's side and ₹5 from the buyer's side) leaves both farmer and buyer meaningfully better off than under the traditional system, while the platform still earns a sustainable, though far smaller, margin than the traditional middleman.

Sample Walkthrough — Farmer

An agent visits a village and helps a tomato farmer list an upcoming harvest 30 days out, with a photo and a reserve price range set with the agent's guidance. A hotel wins the auction. At harvest, the farmer delivers to the local hub, where produce is graded and consolidated with neighboring farmers' harvests into a single truck. Once delivery is confirmed and the inspection window passes without dispute, the farmer receives payment — a fair price, on a predictable timeline, without needing to read the app or negotiate with a middleman directly.

Sample Walkthrough — Institutional Buyer

A hotel kitchen manager, planning for a weekend of events, posts a bulk order for a fixed quantity of vegetables needed on a specific date. Several nearby farmers, through their agents, together commit enough

combined capacity to fulfill it. The produce arrives graded and consolidated, with a clear inspection window to flag any issue before final payment clears.

Sample Walkthrough — Agent

Managing around 70 farmers, an agent spends the week visiting villages, photographing and listing produce for upcoming auctions (often without reliable signal, syncing later), checking in on farmers whose crops are close to harvest, and tracking commission earned across the group.

Sample Walkthrough — Admin

A buyer flags that a delivered batch of onions doesn't match the grade promised at auction. The admin compares the hub's independent grading photos against the buyer's evidence and, finding a genuine partial shortfall, approves a proportional payout reduction rather than a full return, closing the dispute within the fixed inspection window.